identity

- **Inverse Properties**
  - **Inverse Property of Addition:** For any real number a: $a+(-a)=0$ $-a$ is the **additive inverse** of a
  - **Inverse Property of Multiplication:** For any real number a: $(a\neq 0)$ $a\cdot\frac{1}{a}=1$ $\frac{1}{a}$ is the **multiplicative inverse** of a

- **Properties of Zero**
  - **Multiplication by Zero:** For any real number a,
    $a\cdot 0=0$ $0\cdot a=0$ The product of any number and 0 is 0.
  - **Division of Zero:** For any real number a,
    $\frac{0}{a}=0$ Zero divided by any real number, except itself, is zero.
  - **Division by Zero:** For any real number a, $\frac{a}{0}$ is undefined and $a\div 0$ is undefined. Division by zero is undefined.

Exercises

Review Exercises

Rational and Irrational Numbers

In the following exercises, write as the ratio of two integers.

302. 6

303. -5

304. 2.9

305. 1.8

In the following exercises, determine which of the numbers is rational.

306. $0.42, 0.\overline{3}, 2.56813\ldots$

307. $0.75319\ldots, 0.\overline{16}, 1.95$

In the following exercises, identify whether each given number is rational or irrational.

308. ⓐ $\sqrt{49}$ ⓑ $\sqrt{55}$

309. ⓐ $\sqrt{72}$ ⓑ $\sqrt{64}$

In the following exercises, list the ⓐ whole numbers, ⓑ integers, ⓒ rational numbers, ⓓ irrational numbers, ⓔ real numbers for each set of numbers.

310. $-9, 0, 0.361...., \frac{8}{9}, \sqrt{16}, 9$

311. $-5,\ -2\frac{1}{4},\ -\sqrt{4},\ 0.\overline{25},\ \frac{13}{5},\ 4$

Commutative and Associative Properties

In the following exercises, use the commutative property to rewrite the given expression.

312. $6+4=$____

313. $-14\cdot 5=$____

314. $3n=$____

315. $a+8=$____

In the following exercises, use the associative property to rewrite the given expression.

316. $(13\cdot 5)\cdot 2=$_____

317. $(22+7)+3=$_____

318. $(4+9x)+x=$_____

319. $\frac{1}{2}(22y)=$_____

In the following exercises, evaluate each expression for the given value.

320. If $y=\frac{11}{12}$, evaluate:
ⓐ $y+0.7+(-y)$
ⓑ $y+(-y)+0.7$

321. If $z=-\frac{5}{3}$, evaluate:
ⓐ $z+5.39+(-z)$
ⓑ $z+(-z)+5.39$

322. If $k=65$, evaluate:
ⓐ $\frac{4}{9}\left(\frac{9}{4}k\right)$
ⓑ $\left(\frac{4}{9}\cdot\frac{9}{4}\right)k$

323. If $m=-13$, evaluate:
ⓐ $-\frac{2}{5}\left(\frac{5}{2}m\right)$
ⓑ $\left(-\frac{2}{5}\cdot\frac{5}{2}\right)m$

In the following exercises, simplify using the commutative and associative properties.

324. $6y + 37 + (-6y)$

325. $\frac{1}{4} + \frac{11}{15} + \left(-\frac{1}{4}\right)$

326. $\frac{14}{11} \cdot \frac{35}{9} \cdot \frac{11}{14}$

327. $-18 \cdot 15 \cdot \frac{2}{9}$

328. $\left(\frac{7}{12} + \frac{4}{5}\right) + \frac{1}{5}$

329. $(3.98d + 0.75d) + 1.25d$

330. $-12\,(4m)$

331. $30\left(\frac{5}{6}q\right)$

332. $11x + 8y + 16x + 15y$

333. $52m + (-20n) + (-18m) + (-5n)$

Distributive Property

In the following exercises, simplify using the distributive property.

334. $7(x + 9)$

335. $9(u - 4)$

336. $-3(6m - 1)$

337. $-8(-7a - 12)$

338. $\frac{1}{3}(15n - 6)$

339. $(y + 10) \cdot p$

340. $(a - 4) - (6a + 9)$

341. $4(x + 3) - 8(x - 7)$

In the following exercises, evaluate using the distributive property.

342. If $u = 2$, evaluate
(a) $3(8u + 9)$ and
(b) $3 \cdot 8u + 3 \cdot 9$ to show that
$3(8u + 9) = 3 \cdot 8u + 3 \cdot 9$

343. If $n = \frac{7}{8}$, evaluate
(a) $8\left(n + \frac{1}{4}\right)$ and
(b) $8 \cdot n + 8 \cdot \frac{1}{4}$ to show that
$8\left(n + \frac{1}{4}\right) = 8 \cdot n + 8 \cdot \frac{1}{4}$

344. If $d = 14$, evaluate
(a) $-100\,(0.1d + 0.35)$ and
(b) $-100 \cdot (0.1d) + (-100)\,(0.35)$ to show that
$-100\,(0.1d + 0.35) = -100 \cdot (0.1d) + (-100)\,(0.35)$

345. If $y = -18$, evaluate
(a) $-(y - 18)$ and
(b) $-y + 18$ to show that $-(y - 18) = -y + 18$

Properties of Identities, Inverses, and Zero

In the following exercises, identify whether each example is using the identity property of addition or multiplication.

346. $-35(1) = -35$

347. $29 + 0 = 29$

348. $(6x + 0) + 4x = 6x + 4x$

349. $9 \cdot 1 + (-3) = 9 + (-3)$

In the following exercises, find the additive inverse.

350. -32

351. 19.4

352. $\frac{3}{5}$

353. $-\frac{7}{15}$

In the following exercises, find the multiplicative inverse.

354. $\frac{9}{2}$

355. -5

356. $\frac{1}{10}$

357. $-\frac{4}{9}$

In the following exercises, simplify.

358. $83 \cdot 0$

359. $\frac{0}{9}$

360. $\frac{5}{0}$

361. $0 \div \frac{2}{3}$

362. $43 + 39 + (-43)$

363. $(n + 6.75) + 0.25$

364. $\frac{5}{13} \cdot 57 \cdot \frac{13}{5}$

365. $\frac{1}{6} \cdot 17 \cdot 12$

366. $\frac{2}{3} \cdot 28 \cdot \frac{3}{7}$

367. $9(6x - 11) + 15$

Systems of Measurement

In the following exercises, convert between U.S. units. Round to the nearest tenth.

368. A floral arbor is 7 feet tall. Convert the height to inches.

369. A picture frame is 42 inches wide. Convert the width to feet.

370. Kelly is 5 feet 4 inches tall. Convert her height to inches.

371. A playground is 45 feet wide. Convert the width to yards.

372. The height of Mount Shasta is 14,179 feet. Convert the height to miles.

373. Shamu weighs 4.5 tons. Convert the weight to pounds.

374. The play lasted $1\frac{3}{4}$ hours. Convert the time to minutes.

375. How many tablespoons are in a quart?

376. Naomi's baby weighed 5 pounds 14 ounces at birth. Convert the weight to ounces.

377. Trinh needs 30 cups of paint for her class art project. Convert the volume to gallons.

In the following exercises, solve, and state your answer in mixed units.

378. John caught 4 lobsters. The weights of the lobsters were 1 pound 9 ounces, 1 pound 12 ounces, 4 pounds 2 ounces, and 2 pounds 15 ounces. What was the total weight of the lobsters?

379. Every day last week, Pedro recorded the amount of time he spent reading. He read for 50, 25, 83, 45, 32, 60, and 135 minutes. How much time, in hours and minutes, did Pedro spend reading?

380. Fouad is 6 feet 2 inches tall. If he stands on a rung of a ladder 8 feet 10 inches high, how high off the ground is the top of Fouad's head?

381. Dalila wants to make pillow covers. Each cover takes 30 inches of fabric. How many yards and inches of fabric does she need for 4 pillow covers?

In the following exercises, convert between metric units.

382. Donna is 1.7 meters tall. Convert her height to centimeters.

383. Mount Everest is 8,850 meters tall. Convert the height to kilometers.

384. One cup of yogurt contains 488 milligrams of calcium. Convert this to grams.

385. One cup of yogurt contains 13 grams of protein. Convert this to milligrams.

386. Sergio weighed 2.9 kilograms at birth. Convert this to grams.

387. A bottle of water contained 650 milliliters. Convert this to liters.

In the following exercises, solve.

388. Minh is 2 meters tall. His daughter is 88 centimeters tall. How much taller, in meters, is Minh than his daughter?

389. Selma had a 1-liter bottle of water. If she drank 145 milliliters, how much water, in milliliters, was left in the bottle?

390. One serving of cranberry juice contains 30 grams of sugar. How many kilograms of sugar are in 30 servings of cranberry juice?

391. One ounce of tofu provides 2 grams of protein. How many milligrams of protein are provided by 5 ounces of tofu?

In the following exercises, convert between U.S. and metric units. Round to the nearest tenth.

392. Majid is 69 inches tall. Convert his height to centimeters.

393. A college basketball court is 84 feet long. Convert this length to meters.

394. Caroline walked 2.5 kilometers. Convert this length to miles.

395. Lucas weighs 78 kilograms. Convert his weight to pounds.

396. Steve's car holds 55 liters of gas. Convert this to gallons.

397. A box of books weighs 25 pounds. Convert this weight to kilograms.

In the following exercises, convert the Fahrenheit temperatures to degrees Celsius. Round to the nearest tenth.

398. $95°\text{F}$

399. $23°\text{F}$

400. $20°\text{F}$

401. $64°\text{F}$

In the following exercises, convert the Celsius temperatures to degrees Fahrenheit. Round to the nearest tenth.

402. $30°\text{C}$

403. $-5°\text{C}$

404. $-12°\text{C}$

405. $24°\text{C}$

Practice Test

406. For the numbers $0.18349\ldots,\ 0.\overline{2},\ 1.67$, list the (a) rational numbers and (b) irrational numbers.

407. Is $\sqrt{144}$ rational or irrational?

408. From the numbers $-4,\ -1\frac{1}{2},\ 0,\ \frac{5}{8},\ \sqrt{2},\ 7$, which are (a) integers (b) rational (c) irrational (d) real numbers?

409. Rewrite using the commutative property: $x \cdot 14 =$ _____

410. Rewrite the expression using the associative property: $(y + 6) + 3 =$ _____

411. Rewrite the expression using the associative property: $(8 \cdot 2) \cdot 5 =$ _____

412. Evaluate $\frac{3}{16}\left(\frac{16}{3}n\right)$ when $n = 42$.

413. For the number $\frac{2}{5}$ find the (a) additive inverse (b) multiplicative inverse.

In the following exercises, simplify the given expression.

414. $\frac{3}{4}(-29)\left(\frac{4}{3}\right)$

415. $-3 + 15y + 3$

416. $(1.27q + 0.25q) + 0.75q$

417. $\left(\frac{8}{15} + \frac{2}{9}\right) + \frac{7}{9}$

418. $-18\left(\frac{3}{2}n\right)$

419. $14y + (-6z) + 16y + 2z$

420. $9\,(q + 9)$

421. $6(5x - 4)$

422. $-10\,(0.4n + 0.7)$

423. $\frac{1}{4}(8a + 12)$

424. $m(n + 2)$

425. $8\,(6p - 1) + 2\,(9p + 3)$

426. $(12a + 4) - (9a + 6)$

427. $\frac{0}{8}$

428. $\frac{4.5}{0}$

429. $0 \div \left(\frac{2}{3}\right)$

In the following exercises, solve using the appropriate unit conversions.

430. Azize walked $4\frac{1}{2}$ miles. Convert this distance to feet. (1 mile = 5,280 feet).

431. One cup of milk contains 276 milligrams of calcium. Convert this to grams. (1 milligram = 0.001 gram)

432. Larry had 5 phone customer phone calls yesterday. The calls lasted 28, 44, 9, 75, and 55 minutes. How much time, in hours and minutes, did Larry spend on the phone? (1 hour = 60 minutes)

433. Janice ran 15 kilometers. Convert this distance to miles. Round to the nearest hundredth of a mile. (1 mile = 1.61 kilometers)

434. Yolie is 63 inches tall. Convert her height to centimeters. Round to the nearest centimeter. (1 inch = 2.54 centimeters)

435. Use the formula $F = \frac{9}{5}C + 32$ to convert 35°C to degrees F